

ECE 222 Homework #2

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Do the following problems (in bold):

1) Concepts of line, surface, and volume charge density

4.1, 4.2, 4.4, 4.5, 4.8(a), 4.9(a). These help you become familiar with the concepts of line, surface, and volume charge density, as well as practice volume integrals in various coordinates, which you will regularly use in the rest of the class.

2) Calculation of electric field based on Coulomb’s law and the principle of superposition

4.12, 4.16, 4.17, 4.19. Each of these will require you to draw the distribution of charges that are the sources of electric field, the electric field vector that stems from each source, and add them up

Useful integral for 4.16:

$$\int_{-L/2}^{L/2} \frac{dz_0}{(r^2 + z_0^2)^{3/2}} = \frac{1}{r^2} \frac{L}{\sqrt{r^2 + (L/2)^2}}$$

3) Calculation of electric field based on integral form of Gauss’s law

3.50, 4.22, 4.25, 4.26, 4.27, 4.29. Coulomb’s law is often not the best way to calculate the electric field stemming from a given charge distribution. In cases where the charge distribution has a clear symmetry (cylinder, sphere, plane, etc.), the use of Gauss’s law (which, in its integral form, equates the surface integral of $\vec{D}$ through an appropriately chosen surface with the volume integral of charge density ρ_v) is the optimal route to calculate electric field everywhere.

*4.1 A cube 2 m on a side is located in the first octant in a Cartesian coordinate system with one of its corners at the origin. Find the total charge contained in the cube if the charge density is given by $\rho_v = xy^2e^{-2z}$ (mC/m³).

4.2 Find the total charge contained in a cylindrical volume defined by $r \leq 2$ m and $0 \leq z \leq 3$ m if $\rho_v = 20rz$ (mC/m³).

4.4 If the line charge density is given by $\rho_l = 24y^2$ (mC/m), find the total charge distributed on the y axis from $y = -5$ to $y = 5$.

4.5 Find the total charge on a circular disk defined by $r \leq a$ and $z = 0$ if:

- a. $\rho_s = \rho_{s0} \cos \phi$ (C/m²)
- b. $\rho_s = \rho_{s0} \sin^2 \phi$ (C/m²)
- c. $\rho_s = \rho_{s0} e^{-r}$ (C/m²)
- d. $\rho_s = \rho_{s0} e^{-r} \sin^2 \phi$ (C/m²)

where ρ_{s0} is a constant.

4.8 An electron beam shaped like a circular cylinder of radius r_0 carries a charge density given by

$$\rho_v = \left(\frac{-\rho_0}{1+r^2} \right) \text{ (C/m}^3\text{)}$$

where ρ_0 is a positive constant and the beam's axis is coincident with the z axis.

- a. Determine the total charge contained in length L of the beam.
- b. If the electrons are moving in the $+z$ direction with uniform speed u , determine the magnitude and direction of the current crossing the z -plane.

4.9 A circular beam of charge of radius a consists of electrons moving with a constant speed u along the $+z$ direction. The beam's axis is coincident with the z axis and the electron charge density is given by

$$\rho_v = -cr^2 \quad (\text{C/m}^3)$$

where c is a constant and r is the radial distance from the axis of the beam.

- *a. Determine the charge density per unit length.
- b. Determine the current crossing the z -plane.

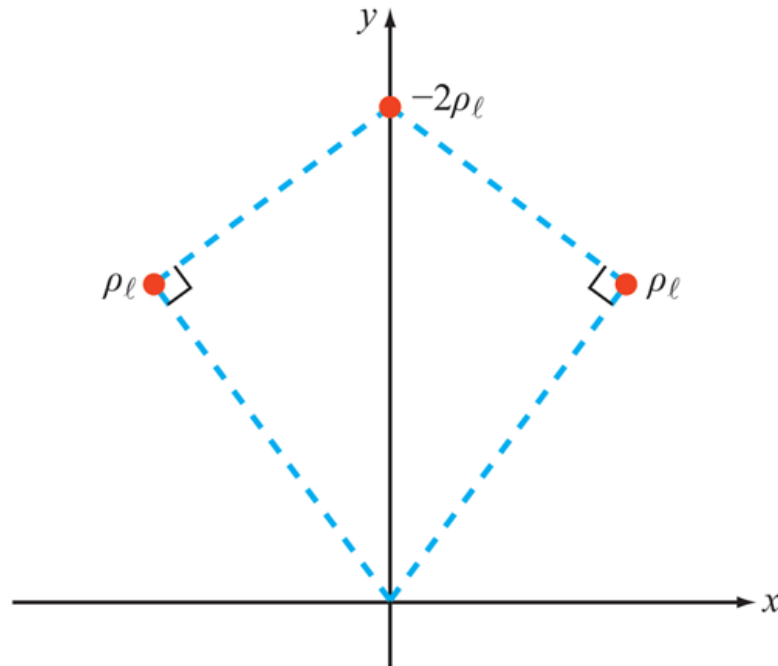
- 4.12 Three point charges, each with $q = 3 \text{ nC}$, are located at the corners of a triangle in the x - y plane, with one corner at the origin, another at $(2 \text{ cm}, 0, 0)$, and the third at $(0, 2 \text{ cm}, 0)$. Find the force acting on the charge located at the origin.
- 4.16 A line of charge with uniform density ρ_l extends between $z = -L/2$ and $z = L/2$ along the z axis. Apply Coulomb's law to obtain an expression for the electric field at any point $P(r, \phi, 0)$ on the x - y plane. Show that your result reduces to the expression given by Eq. (4.33) as the length L is extended to infinity.
- *4.17 Repeat Example 4-5 for the circular disk of charge of radius a , but in the present case, assume the surface charge density to vary with r as

$$\rho_s = \rho_{s0} r^2 \quad (\text{C/m}^2)$$

where ρ_{s0} is a constant.

- *4.19 Three infinite lines of charge, all parallel to the z axis, are located at the three corners of the kite-shaped arrangement shown in Fig. P4.19. If the two right triangles are symmetrical and of equal corresponding sides, show that the electric field is zero at the origin.

Figure P4.19 Kite-shaped arrangement of line charges for Problem 4.19.



See next page for Eq. (4.33) and Example 4-5 for problems 4.16 and 4.17

(4.33)

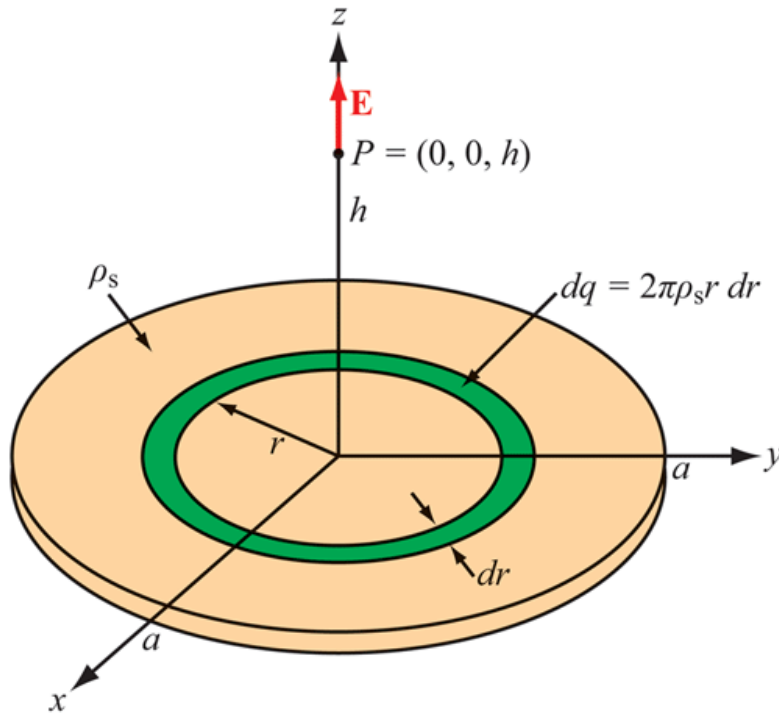
$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon_0} = \hat{\mathbf{r}} \frac{D_r}{\epsilon_0} = \hat{\mathbf{r}} \frac{\rho_\ell}{2\pi\epsilon_0 r}.$$

(infinite line charge)

Example 4-5: Electric Field of a Circular Disk of Charge

Find the electric field at point P with Cartesian coordinates $(0, 0, h)$ due to a circular disk of radius a and uniform charge density ρ_s residing in the x - y plane (Fig. 4-7). Also, evaluate $\mathbf{E}$ due to an infinite sheet of charge density ρ_s by letting $a \rightarrow \infty$.

Figure 4-7 Circular disk of charge with surface charge density ρ_s . The electric field at $P = (0, 0, h)$ points along the z direction (Example 4-5).



Solution: Building on the expression obtained in [Example 4-4](#) for the on-axis electric field due to a circular ring of charge, we can determine the field due to the circular disk by treating the disk as a set of concentric rings. A ring of radius r and width dr has an area $ds = 2\pi r dr$ and contains charge $dq = \rho_s ds = 2\pi\rho_s r dr$. Upon using this expression in [Eq. \(4.23\)](#) and also replacing b with r , we obtain the following expression for the field due to the ring:

$$d\mathbf{E} = \hat{\mathbf{z}} \frac{h}{4\pi\epsilon_0(r^2 + h^2)^{3/2}} (2\pi\rho_s r dr).$$

The total field at P is obtained by integrating the expression over the limits $r = 0$ to $r = a$:

(4.24)

$$\mathbf{E} = \hat{\mathbf{z}} \frac{\rho_s h}{2\epsilon_0} \int_0^a \frac{r dr}{(r^2 + h^2)^{3/2}} = \pm \hat{\mathbf{z}} \frac{\rho_s}{2\epsilon_0} \left[1 - \frac{|h|}{\sqrt{a^2 + h^2}} \right],$$

with the plus sign for $h > 0$ (P above the disk) and the minus sign when $h < 0$ (P below the disk).

For an infinite sheet of charge with $a = \infty$,

(4.25)

$$\mathbf{E} = \pm \hat{\mathbf{z}} \frac{\rho_s}{2\epsilon_0} .$$

(infinite sheet of charge)

We note that for an infinite sheet of charge $\mathbf{E}$ is the same at all points above the x - y plane, and a similar statement applies for points below the x - y plane.

- *3.50 A vector field $\mathbf{D} = \hat{\mathbf{r}}r^3$ exists in the region between two concentric cylindrical surfaces defined by $r = 1$ and $r = 2$ with both cylinders extending between $z = 0$ and $z = 5$.

Verify the divergence theorem by evaluating the following:

$$\begin{aligned} \text{a. } & \oint_S \mathbf{D} \cdot d\mathbf{s} \\ \text{b. } & \int_v \nabla \cdot \mathbf{D} \, dv \end{aligned}$$

- 4.22 Given the electric flux density

$$\mathbf{D} = \hat{\mathbf{x}}2(x + y) + \hat{\mathbf{y}}(3x - 2y) \quad (\text{C/m}^2),$$

determine

- ρ_v by applying Eq. (4.26) $\square$,
- the total charge Q enclosed in a cube 2 m on a side, located in the first octant with three of its sides coincident with the x , y , and z axes and one of its corners at the origin, and
- the total charge Q in the cube, obtained by applying Eq. (4.29) $\square$.

Eq. (4.26):

(4.26)

$$\nabla \cdot \mathbf{D} = \rho_v,$$

(differential form of Gauss's law)

Eq. (4.29):

(4.29)

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q.$$

(integral form of Gauss's law)

- *4.25 The electric flux density inside a dielectric sphere of radius a centered at the origin is given by

$$\mathbf{D} = \hat{\mathbf{R}}\rho_0 R \quad (\text{C/m}^2)$$

where ρ_0 is a constant. Find the total charge inside the sphere.

- 4.26 In a certain region of space, the charge density is given in cylindrical coordinates by the function

$$\rho_v = 5re^{-r} \quad (\text{C/m}^3).$$

Apply Gauss's law to find $\mathbf{D}$.

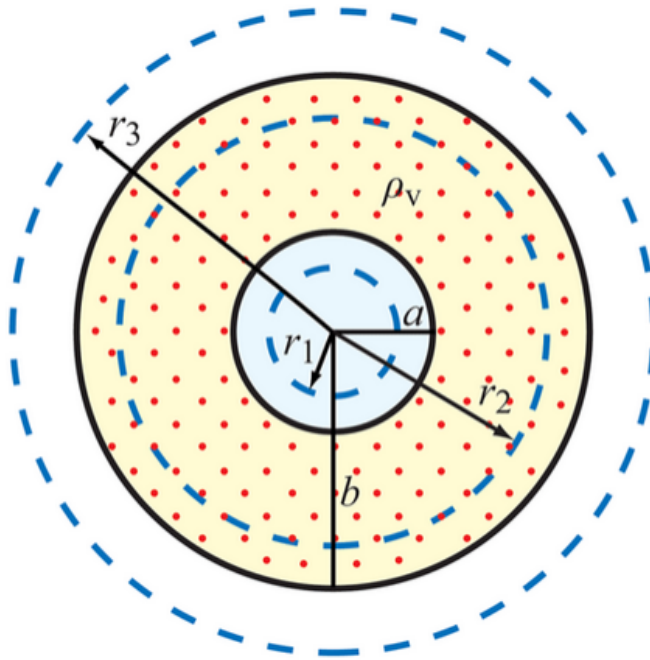
- *4.27 An infinitely long cylindrical shell extending between $r = 1 \text{ m}$ and $r = 3 \text{ m}$ contains a uniform charge density ρ_{v0} . Apply Gauss's law to find $\mathbf{D}$ in all regions.

- 4.29** A spherical shell with outer radius b surrounds a charge-free cavity of radius $a < b$ (Fig. P4.29). If the shell contains a charge density given by

$$\rho_v = -\frac{\rho_{v0}}{R^2}, \quad a \leq R \leq b,$$

where ρ_{v0} is a positive constant, determine $\mathbf{D}$ in all regions.

Figure P4.29 Problem 4.29.



Solutions to starred problems:

4.1 $Q = 2.62 \text{ (nC)}$

4.17 $\mathbf{E} = \hat{\mathbf{z}} (\rho_{s0} h / 2\epsilon_0) [\sqrt{a^2 + h^2} + h^2 / \sqrt{a^2 + h^2} - 2h]$

4.19 $\mathbf{E} = 0$

$$3.50 \blacksquare \quad (\text{a}) \oint \mathbf{D} \cdot d\mathbf{s} = 150\pi$$

$$(\text{b}) \iiint \nabla \cdot \mathbf{D} dv = 150\pi$$

$$4.25 \blacksquare \quad Q = 4\pi\rho_0 a^3 \text{ (C)}$$

$$4.27 \blacksquare \quad \mathbf{D} = \hat{\mathbf{r}} \frac{\rho_{v0}(r^2 - 1)}{2r}, \quad 1 \leq r \leq 3 \text{ m}$$

$$\mathbf{D} = \hat{\mathbf{r}} D_r = \hat{\mathbf{r}} \frac{4\rho_{v0}}{r}, \quad r \geq 3 \text{ m}$$